3–9 (a) Formulate mesh-current equations for the circuit in Figure P3–9.
(b) Use these equations to find v_x and i_x.

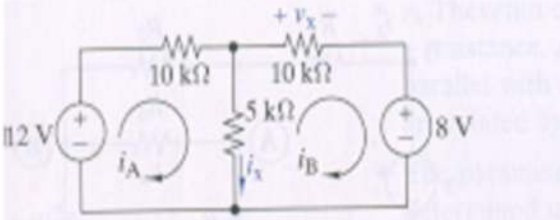

FIGURE P3–9

3–11 (a) Formulate mesh-current equations for the circuit in Figure P3–11.
(b) Use these equations to find v_x and i_x.

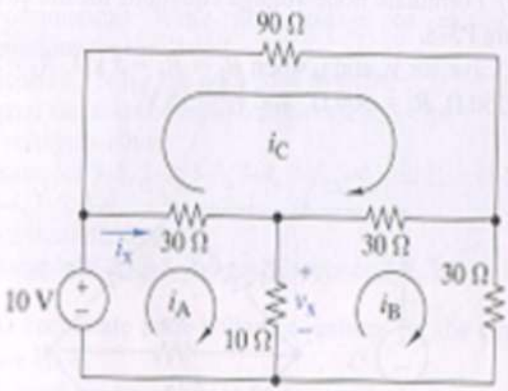

FIGURE P3–11

3–12 (a) Formulate mesh-current equations for the circuit in Figure P3–12.
(b) Solve for v_x and i_x when $R_1 = 200\ \Omega$, $R_2 = 300\ \Omega$, $R_3 = 50\ \Omega$, $R_4 = 250\ \Omega$, $R_5 = 200\ \Omega$, $i_S = 50$ mA, and $v_S = 15$ V.
(c) Find the total power dissipated in the circuit.

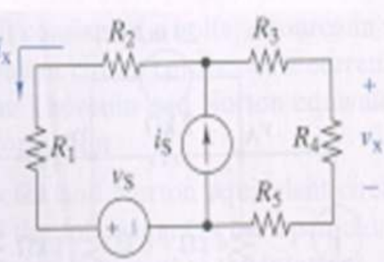

FIGURE P3–12

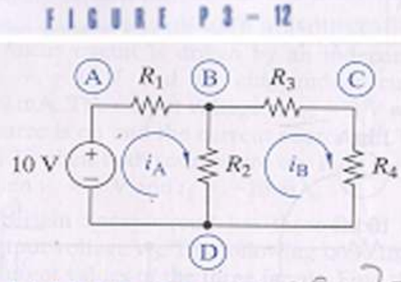

FIGURE P3–17 28 2
56

3–18 In Figure P3–17 all of the resistors are 1 kΩ. The voltage at node A is observed to be $v_A = 8$ V when node C is connected to ground. Find the node voltages v_B and v_D, and the mesh currents i_A and i_B.

3–27 Use the superposition principle to find i_O in Figure P3–27.

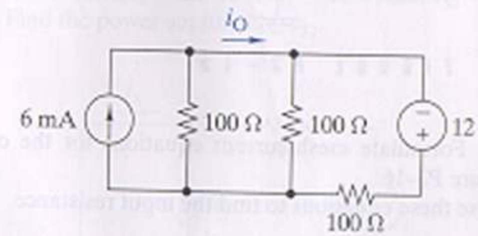

FIGURE P3–27

3–30 Use the superposition principle to find v_O in terms of v_1, v_2, and R in Figure P3–30.

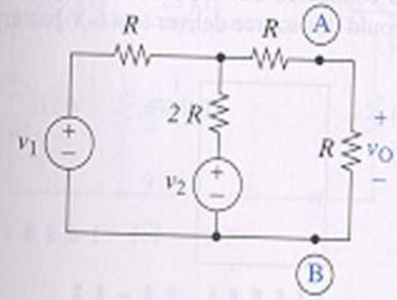

FIGURE P3–30

4–4 Find the voltage gain v_O/v_S and current gain i0/ix Figure P4–4.

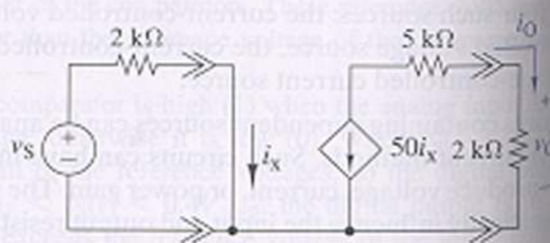

FIGURE P4–4

4–10 Find an expression for the voltage gain v_O/v_S in Figure P4–10.

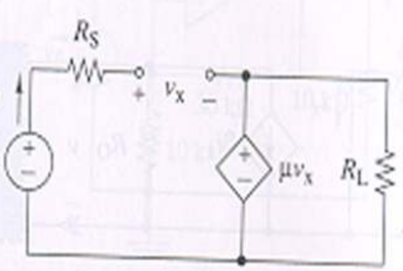

FIGURE P4–10

4–12 Find R_{IN} in Figure P4–12.

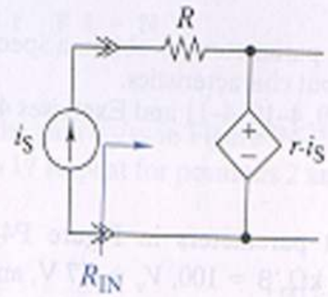

FIGURE P4 – 12

4–14 Find the Thévenin equivalent circuit seen by the load in Figure P4–14.

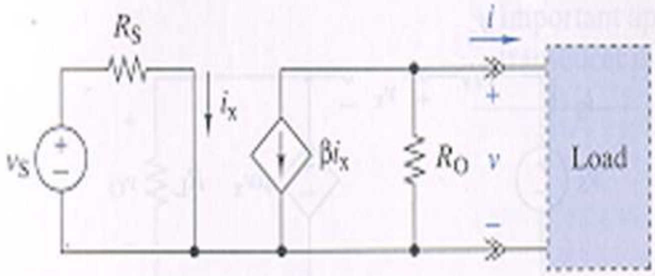

FIGURE P4 – 14